

A Novel Current-Mode Quaternary Multiplier with Indian Vedic Architecture

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Abstract

A multiplier is a fundamental block of any digital signal processing system. Urdhva Tiryakbhyam sutra, from ancient Indian Vedic mathematics, offers regular and hierarchical multiplier architecture. Alternatively, current-mode multi-valued logic results in reduced circuitry and effective utilization of interconnections. In this paper, a new 4×4 current-mode quaternary Vedic multiplier design is proposed. The multiplier has been simulated using 0.18 μm technology with a power supply of 1 V and a unit current step of 1 μA . The proposed multiplier works on simple quaternary arithmetic and has a very low transistor-count. The transient delay of the proposed multiplier also has been improved compared to our previous multiplier. The proposed approach of employing multi-valued logic with Vedic architecture has made the application of multi-valued logic simpler, which can make multi-valued circuits as generalized as binary circuits.

Keywords: Multi-Valued Logic, Current-Mode Circuits, Vedic Multiplier, Urdhva Tiryakbhyam Sutra, Digital Signal Processing

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INTRODUCTION

Digital signal processing (DSP) systems manipulate the signals as a sequence of numbers, and usually require extensive arithmetic computations to perform algorithmic operations such as modulation, transforms, and filtering. Multipliers are the basic functional units in such DSP applications and consume a considerable amount of hardware. A DSP chip needs compact, fast, reliable, and low power consuming multipliers. Performance of multipliers often affects the performance of the entire DSP system. Hence, the efficient design of multiplier is still a popular research subject. High performance multiplier design can be achieved by using efficient adders and a better architecture. In this paper, we have presented a new current-mode quaternary adder, which is then used to design a quaternary multiplier with Vedic architecture.

Multi-Valued Logic

As the name suggests, multi-valued logic (MVL) uses more than two logic levels to

define a signal, such as ternary, quaternary, or even more. The advantages of MVL over binary logic are smaller circuitry, greater speed, reduced number of interconnections, reduced cross-talk noise, and increased data processing capability per unit of chip area. It combines some of the efficiency of analogue signaling with the noise immunity of digital signaling [1].

MVL can be employed in two modes, viz., voltage mode and current mode. In voltage-mode MVL circuits, the operating voltage range is divided into the number of logical values to be represented. In current-mode MVL circuits, integer multiples of a unit current step are assigned the logical levels. Current-mode implementation reduces the number of active devices and wiring complexity compared to the voltage-mode, since the linear sum of outputs can be performed simply by connecting those wires together. Hence current-mode MVL is preferred while designing arithmetic circuits. However, current-mode MVL circuits are

assumed to have smaller minimum driving current, resulting in slower operation compared to binary circuits. Current mirrors, current comparators, and switched current sources are the basic building blocks used to design most of the arithmetic circuits in current-mode MVL such as adders, subtractors, multipliers, etc. [2–3].

Vedic Multiplier

The literal meaning of the word “Veda” in Sanskrit is “knowledge”. Indian intellectuals of ancient times had written these Vedas, which consist of immense knowledge useful to humankind from various aspects of life. These Vedas have made many useful contributions to medical science and mathematics in India. Vedic mathematics techniques were rediscovered in the early twentieth century from ancient Indian scriptures by Shri Bharati Krishna Tirthaji Maharaj. These methods can be applied directly to plain and spherical geometry, trigonometry, conics, calculus, and applied mathematics of various kinds.

Vedic mathematics is a name often heard in India with reference to the techniques for mentally solving mathematical problems. It is based on 16 *sutras* or formulae, each of them having *upa-sutras* or corollaries. Two of these

sutras, *Nikhilam sutra* and *Urdhva Tiryakbhyam sutra*, are useful for multiplication.

Nikhilam sutra, meaning “all from 9 and last from 10”, provides an efficient multiplication algorithm when the operands are very large and close to an integer power of 10. However, both the operands should be either less than or greater than the selected power of 10 [4].

Urdhva Tiryakbhyam sutra, meaning “vertically and crosswise”, is applicable in all cases of multiplication. Figure 1 illustrates the multiplication of two 4-digit numbers, X and Y , using *Urdhva Tiryakbhyam sutra*. The digits on both sides of the line are multiplied, and the product is added with the carry from the previous step. This generates the result and a carry. This carry is added in the next step and thus the process goes on. If more than one line is there in one step, all of the products are added along with the previous carry. In each step, the least significant digit acts as the result digit and all other digits act as a carry for the next step. Initially, the carry is taken to be zero. When employed manually, *Urdhva Tiryakbhyam sutra* gives the final product in a single line.

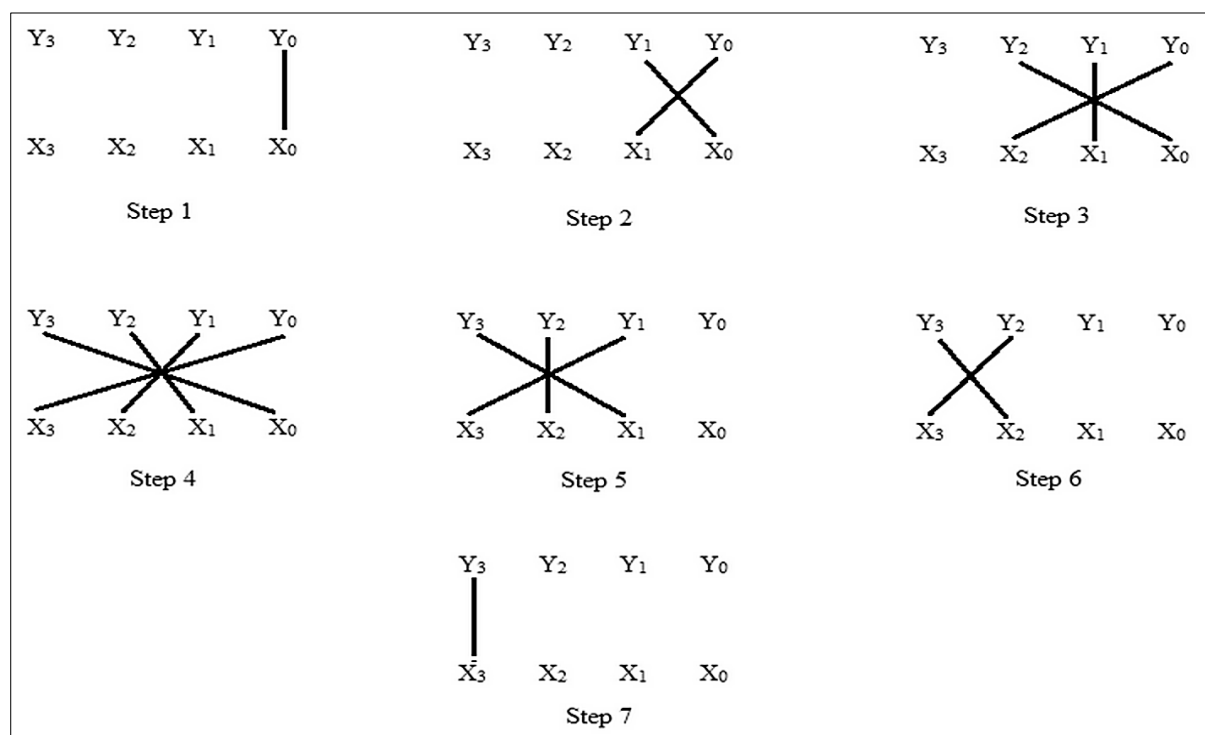


Fig. 1: Line Diagram of 4×4 Multiplication using Urdhva Tiryakbhyam Sutra.

The expressions used for calculating each term in the product, with $C_6P_6P_5P_4P_3P_2P_1P_0$ as the final product, are:

$$\text{Step 1: } C_0P_0 = (X_0 \times Y_0)$$

$$\text{Step 2: } C_1P_1 = (X_0 \times Y_1) + (X_1 \times Y_0) + C_0$$

$$\text{Step 3: } C_2P_2 = (X_0 \times Y_2) + (X_1 \times Y_1) + (X_2 \times Y_0) + C_1$$

$$\text{Step 4: } C_3P_3 = (X_0 \times Y_3) + (X_1 \times Y_2) + (X_2 \times Y_1) + (X_3 \times Y_0) + C_2$$

$$\text{Step 5: } C_4P_4 = (X_1 \times Y_3) + (X_2 \times Y_2) + (X_3 \times Y_1) + C_3$$

$$\text{Step 6: } C_5P_5 = (X_2 \times Y_3) + (X_3 \times Y_2) + C_4$$

$$\text{Step 7: } C_6P_6 = (X_3 \times Y_3) + C_5$$

The expressions above are valid for multiplication using Urdhva Tiryakbhyam *sutra* in any number system. From Figure 1, it is clear that Vedic multiplication is much simpler than the conventional method of multiplication. Vedic multiplication strikes a difference in the actual process of multiplication itself. It enables parallel generation of intermediate products and eliminates unwanted multiplication steps with zeros. Although the number of multiplications done in the Vedic method is equal to that in the conventional method, the Vedic multiplier requires fewer adders compared to the conventional method [5]. Vedic architecture provides chances for hierarchical design where a smaller multiplier is used to design a larger one. This is also helpful for designing a multiplier in VHDL, as it gives effective utilization of the structural method of modeling. A Vedic multiplier requires fewer look-up tables, which reduces the hardware and memory requirement. The hierarchical structure and effective addition of partial products enable large savings of resources when used in ASIC or FPGA for DSP. So, be it ASIC implementation or implementation using hardware description languages, employing Vedic techniques reduces the complexity, execution time, chip area, power consumption, and hardware cost of the multiplier [4–6].

PREVIOUS WORK

Work in MVL

Since no standard circuitry has been established in MVL, different researchers apply their own logics. The research work done in MVL mainly is focused on designing the adders and multipliers. Most of the researchers have used redundant number systems to design the multipliers in MVL. In

[7], Alejandro, et al. presented a detailed survey of circuit implementations of redundant arithmetic algorithms used in MVL circuits. In these number systems, redundancy occurs because contrary to the case of conventional systems, redundant numbers can have more than one representation. Even though redundant algorithms produce significant improvements in performance through the reduction or elimination of carry propagation chains, efficient circuit implementations of these algorithms are difficult to obtain [7].

Circuit complexity has traditionally been one of the drawbacks of MVL implementation [7]. For instance, Ishizuka, et al. proposed a 4×4 current-mode quaternary multiplier using a redundant number system [8]. This resulted in a reduced number of adder stages and, hence, reduced circuitry. However, neither 1×1 nor 2×2 multipliers were designed, nor can this 4×4 multiplier be used to design the multiplier with higher configuration. A redundant number system also increases the mathematical complexity of the design, which makes such circuits less preferred compared to their binary counterparts. Thus, there is a need of MVL circuitry based on simpler working principles with hierarchical structures, in order to make MVL as suitable as binary logic.

Work in Vedic Multiplier

The Vedic multiplication method has been extended to binary logic in the literature. However, the work done is more theoretical and fundamental. Some of the researchers have explained the workings of Nikhilam *sutra* and Urdhva Tiryakbhyam *sutra* [9–11], whereas, some have explained the architectures of Vedic multipliers [12–13]. Most of the Vedic multipliers in the available literature are designed using hardware

description languages [9–13]. In [4], Prabir, et al. claimed to have realized a transistor level design of a Vedic multiplier, but neither any transistor level circuitry nor the transistor-count has been specified. Successful FPGA implementation and testing of a Vedic multiplier in [9–13] showed that the theory of Vedic multiplication could be transformed into a multiplier design. This makes Vedic multiplication a valid method of multiplication for DSP operations.

Researchers working with a Vedic multiplier have implemented Vedic *sutras* as they are, without making any modification at the circuit level. Although this gives some advantages over a conventional multiplier, the work is limited to the architecture level only. To explore fully the advantages of the Vedic

multiplication method, there is a need to design efficient circuit blocks that can be used with Vedic architecture.

PROPOSED 1×1 QUATERNARY MULTIPLIER

In binary logic, the AND gate acts as a 1×1 multiplier. However, in quaternary logic, the actual 1×1 multiplier is required. Table 1 gives the truth table of 1×1 quaternary multiplication. Quaternary multiplication takes place in two steps. The first step is the generation of a partial product, and the second step is the conversion of this partial product into an actual product and a carry. The partial product is calculated as $u=(x \times y)$. The partial product u is converted into an actual product and a carry as $u=(4c+p)$.

Table 1: Truth Table for 1×1 Quaternary Multiplication.

Sr. No.	Input (x)	Input (y)	Partial Product (u)	Carry (c)	Product (p)
1.	0	0	0	0	0
2.	0	1	0	0	0
3.	0	2	0	0	0
4.	0	3	0	0	0
5.	1	0	0	0	0
6.	1	1	1	0	1
7.	1	2	2	0	2
8.	1	3	3	0	3
9.	2	0	0	0	0
10.	2	1	2	0	2
11.	2	2	4	1	0
12.	2	3	6	1	2
13.	3	0	0	0	0
14.	3	1	3	0	3
15.	3	2	6	1	2
16.	3	3	9	2	1

In this paper, we have used a current-mode multiplier-divider [14] as a partial product generator (PPG). Figure 2 shows the PPG, where I_1 , I_2 and I_3 are input currents and I_{out} is the output current. V_C is the cascode bias voltage. The aspect ratio of transistors in the upper row is nine times its original value “M”. Total current drawn by the left half of the circuit is:

$$I_{left} = 2\sqrt{I_1 \times I_2} \quad (1)$$

The right half of the circuit is analogous to the

left half. Hence, I_{right} can be expressed as

$$I_{right} = 2\sqrt{I_3 \times I_{out}} \quad (2)$$

Since I_{right} is the mirror image of I_{left} ,

$$I_{left} = I_{right} \quad (3)$$

Solving (1)–(3), we obtain:

$$I_{out} = (I_1 \times I_2) / I_3$$

By keeping I_3 constant at logic “1”, we have used this multiplier-divider as PPG, whose output is:

$$I_{out} = (I_1 \times I_2)$$

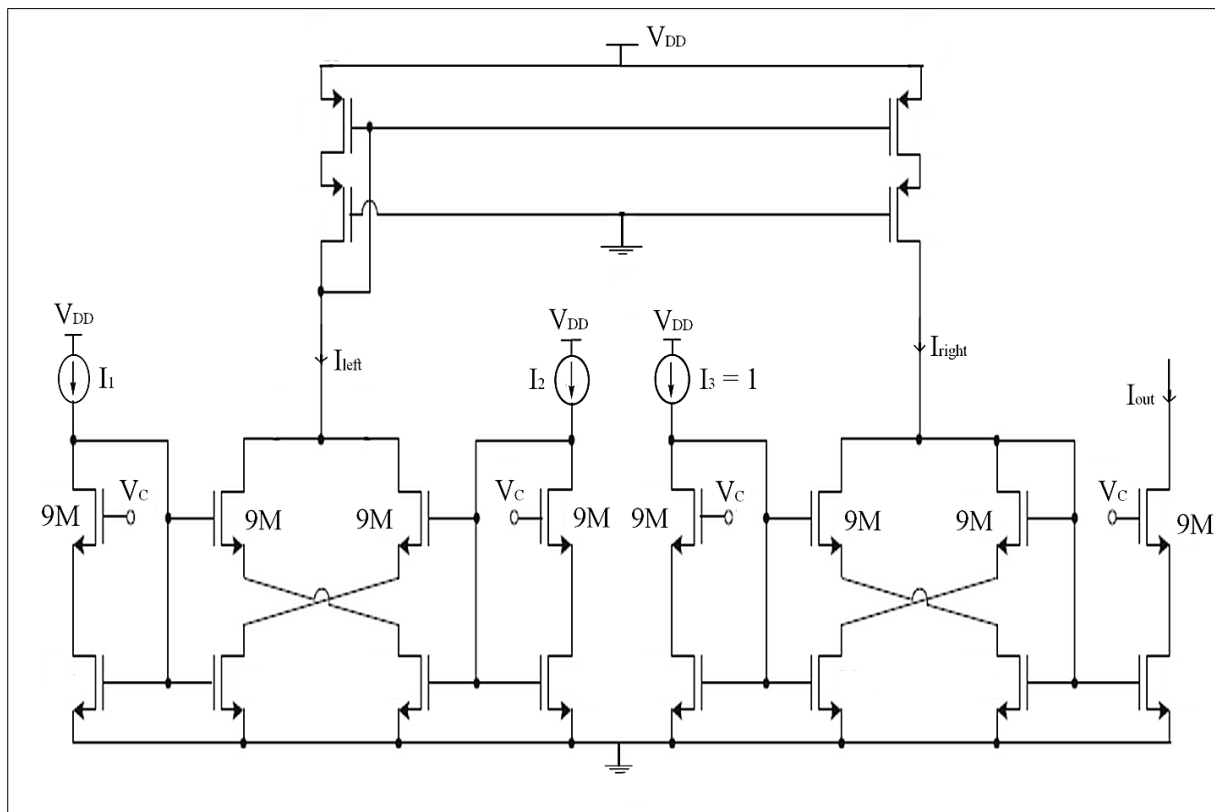


Fig. 2: Partial Product Generator (PPG).

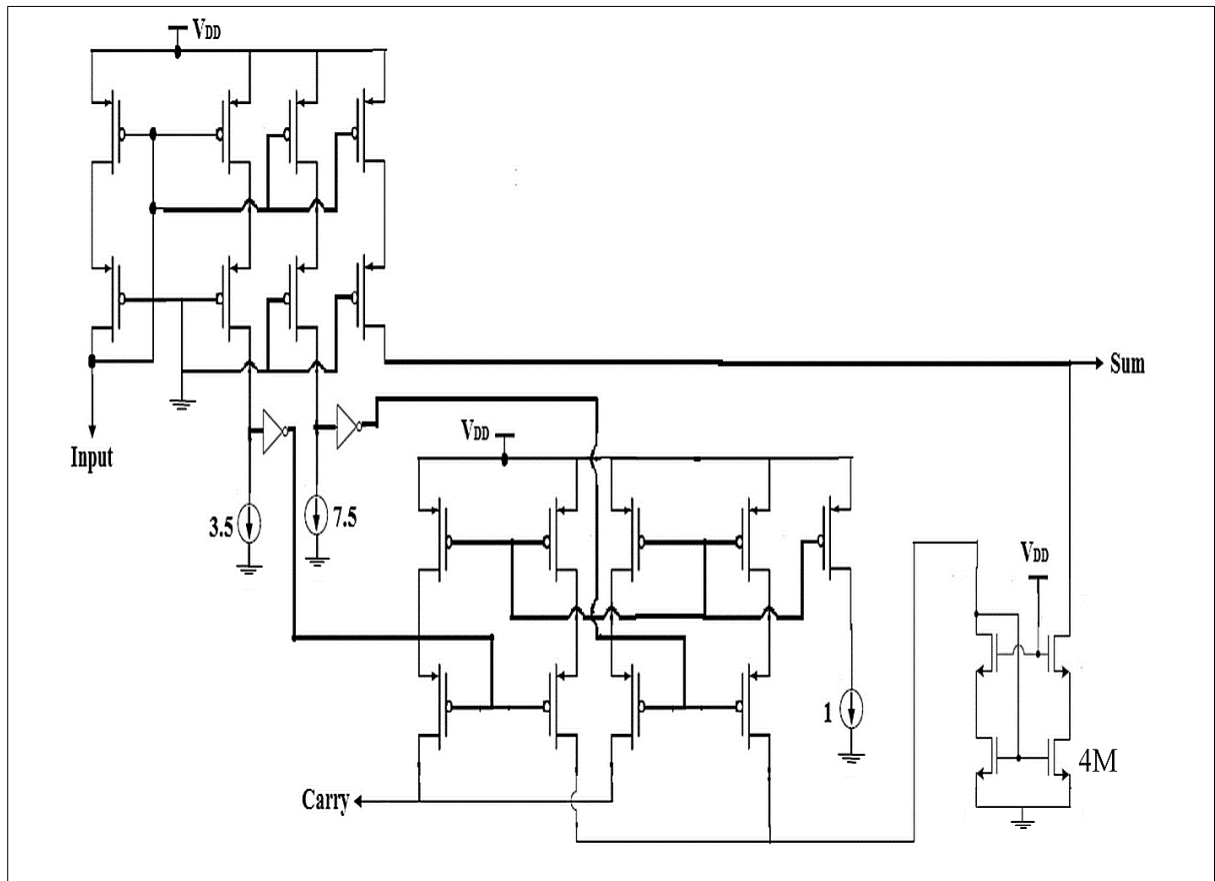


Fig. 3: Proposed QFA.

I_{out} is the partial product, which should be converted into a proper quaternary number using the quaternary full adder (QFA). We have designed a single block for QFA by combining high-swing cascode current mirror (HCCM) topology from [15] with the logic of a rail-comparator from [16], as shown in Figure 3. It consists of two PMOS HCCM mirrors and one NMOS HCCM mirror. The logic of a rail-comparator is incorporated with the upper PMOS mirror. The lower PMOS mirror produces four copies of current equal to logic “1”. The NMOS mirror connected at the output is designed in such a way that it draws current equal to four times its input current. This is done by changing the aspect ratio of the output transistor to four times its original value. Input current of QFA is drawn from the upper PMOS mirror, which produces its three mirror images, two of which are compared with the reference currents of logic “3.5” and “7.5”. If the input is greater than the reference current, V_{DD} will be produced at the corresponding output branch, which will be converted into GND voltage through the CMOS inverter. This will turn ON the corresponding two PMOSes connected with the output of that CMOS inverter. Thus, two copies of logic “1” will be passed as the outputs of the lower PMOS mirror. One of them will be given as the carry, and another

will be given to the NMOS mirror, which will subtract “4” from the third mirror image of input current of QFA. Figure 4 shows the block diagram of the proposed 1×1 quaternary multiplier, formed by connecting PPG and QFA together.

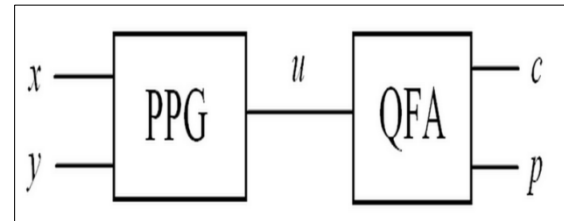


Fig. 4: Block Diagram of 1×1 Quaternary Multiplier.

PROPOSED 4×4 QUATERNARY VEDIC MULTIPLIER

The architecture of the proposed multiplier is based on Urdhva Tiryakbhyam *sutra* since it is applicable to all cases of multiplication. From Figures 5 and 6, it can be concluded that the application of the current-mode MVL reduces the number of adders in the Vedic multiplier as the multiplier configuration continues to increase. Moreover, the 4×4 quaternary multiplier in Figure 6(b) is equivalent to an 8×8 binary multiplier, which shows a great reduction in the circuitry [17].

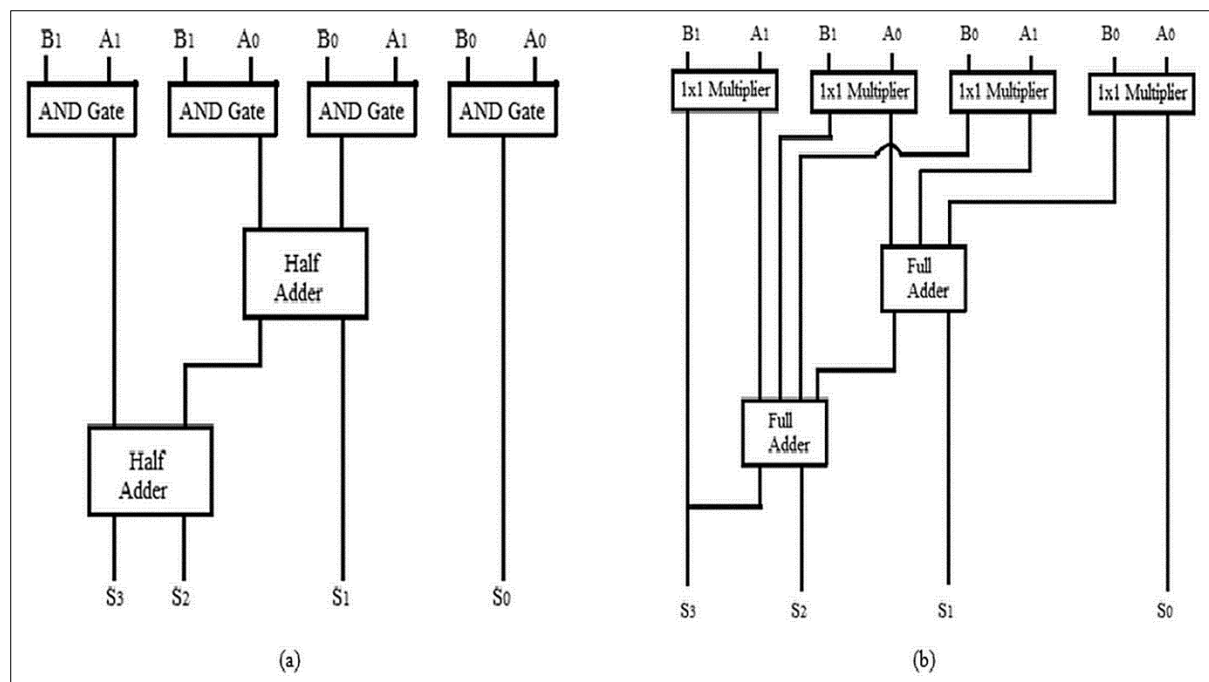


Fig. 5: Architecture of 2×2 Vedic Multiplier in: (a) Binary Logic, and (b) Current-Mode Quaternary Logic [17].

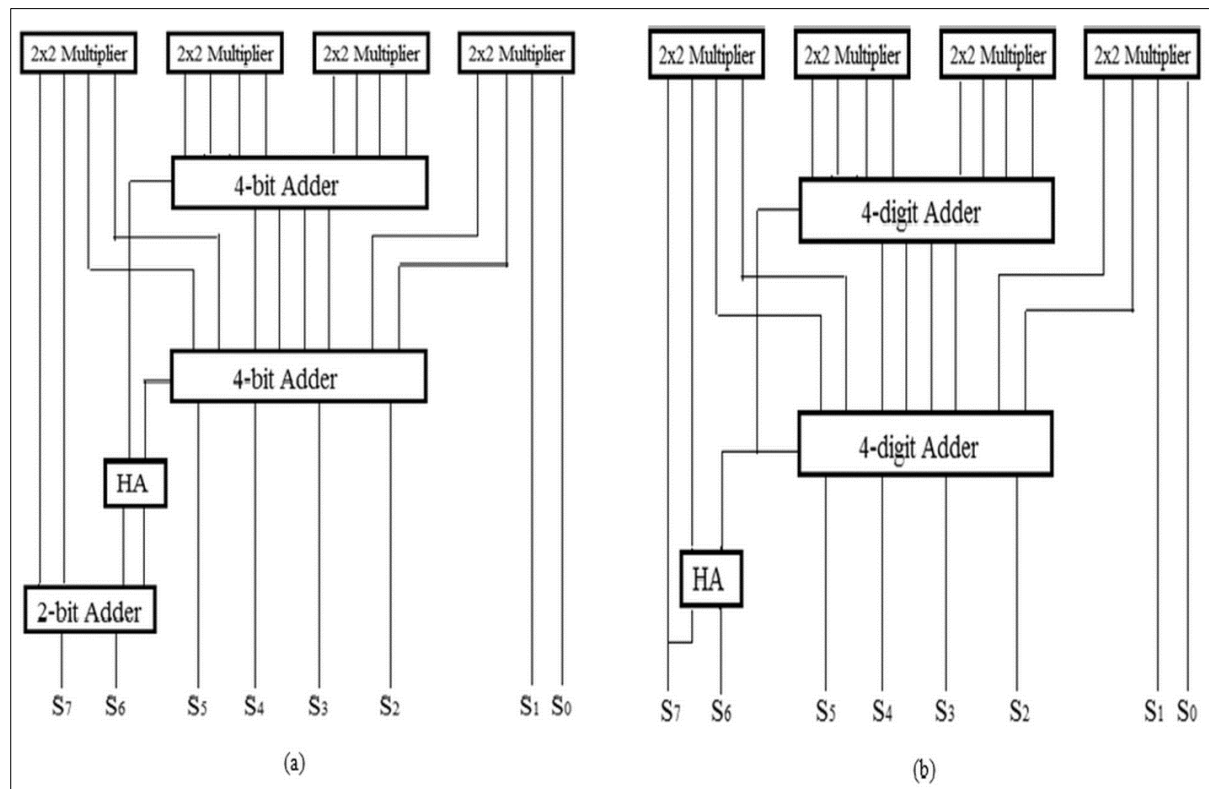


Fig. 6: Architecture of 4×4 Vedic Multiplier in: (a) Binary Logic, and (b) Current-Mode Quaternary Logic [17].

As shown in Figure 5(b), full adder (FA) is required to design a 2×2 quaternary current-mode Vedic multiplier. In the proposed approach, QFA and FA have equivalent logic functions of quaternary addition. However, QFA is used for converting the partial product into the final product and a carry of a 1×1 multiplier; whereas, FA is used for actual quaternary addition. The maximum possible input to QFA is “9” since it is associated with PPG; whereas, as per the proposed design, the maximum possible input to FA is “7”. Hence, FA is designed similar to QFA but without the comparator branch with a reference current of logic “7.5”.

A 4-digit adder and a half adder (HA) are required for the 4×4 quaternary Vedic multiplier in Figure 6(b). We have used a simple ripple carry adder structure to design the 4-digit adder. FA and HA are differentiated on the basis of the total number of inputs. We have used the topology of the adder such that the linear sum of all the adder inputs is considered a single input of the adder. Hence, we have used the same block of FA as HA.

RESULTS, ANALYSIS, AND COMPARISON

The proposed multiplier has been simulated using TSPICE in $0.18 \mu\text{m}$ technology with a supply voltage of 1 V. The unit current step is taken as $1 \mu\text{A}$, i.e., logic levels 0, 1, 2, and 3 are assigned to $0 \mu\text{A}$, $1 \mu\text{A}$, $2 \mu\text{A}$, and $3 \mu\text{A}$, respectively. Cascode bias voltage V_C of PPG is 0.5 V. To study the merits and demerits of the proposed multiplier, the size, speed, and power performances have been analyzed and compared with those in the literature. Although every design has its own logic, we have tried to make the comparisons as fair as possible.

Comparison of Transistor-count

Table 2 gives the comparison of proposed FA with some quaternary current-mode full adders in the literature. While determining the transistor-count of [18], we have not considered the transistors used for flip-flops. Table 2 shows that the proposed FA requires substantially fewer transistors.

Table 2: Comparison of Transistor-count of Proposed FA.

	Transistor-count	% Reduction
Proposed	17	-
[17]	23	26.08%
[18]	63	73.01%

The proposed 4×4 quaternary multiplier is compared with multipliers in the literature in terms of transistor-count, as shown in Table 3. Reference [17] (Shende A. S., et al.), is our previous multiplier design, which proposes a 4×4 quaternary current-mode Vedic multiplier. An 8×8 current-mode binary multiplier was proposed in [19], and [20] proposed an 8×8 binary array multiplier. The proposed multiplier requires significantly fewer transistors compared to other multipliers.

Table 3: Comparison of Transistor-count of Proposed 4×4 Multiplier.

	Transistor-count	% Reduction
Proposed	1077	-
[17]	1271	15.26%
[19]	2510	57.09%
[20]	1952	44.82%

Comparative Analysis of Delay and Power Consumption

In binary circuits, the delay is usually measured between 50% of input transition and a corresponding 50% of output transition. However, this definition of delay is not acceptable for MVL circuits since the linearity in the output is not like binary circuits because of more than two logic levels. No standard definition of delay was found in the available literature. Hence, we have used our own definition of delay for analysis purposes. We calculated the transient delay as “the time required for the output to reach a value equal to ‘(final value of output) ± (10% of unit step current)’ after the input becomes stable”. We applied 48 different variations at different inputs of the proposed multiplier, which gave us 206 readings of transient delay at the outputs and 48 readings of power consumption. A complete analysis of transient delay and power consumption of the proposed multiplier is shown in Table 4, which also gives a comparison of the proposed multiplier with our previous multiplier in [17]. Delay and power consumption of [17] were calculated for similar combinations and variations as those of the proposed multiplier using the same definition. Supply voltage, design technology, and unit current step were also similar.

Table 4: Comparative Analysis of Transient Delay and Power Consumption of Proposed Multiplier.

Parameter	Proposed	[17]
Best-case Delay	1.66 ns	16.42 ns
Worst-case Delay	119.07 ns	174.77 ns
Difference Between Best-case and Worst-case Delays	117.41 ns	158.35 ns
Average of Best-case and Worst-case Delays	60.36 ns	95.59 ns
Average of All Readings of Delay	59.07 ns	64.63 ns
Average of Best 30 Readings of Delay	15.20 ns	25.25 ns
Average of Worst 30 Readings of Delay	102.19 ns	116.44 ns
Best-case Power Consumption	7.88 mW	0.50 mW
Worst-case Power Consumption	10.90 mW	0.70 mW
Difference Between Best-case and Worst-case Power Consumptions	3.02 mW	0.20 mW
Average of Best-case and Worst-case Power Consumptions	9.39 mW	0.60 mW
Average of All Readings of Power Consumption	9.08 mW	0.59 mW
Power Consumption for Best-case Delay	8.36 mW	0.52 mW
Power Consumption for Worst-case Delay	8.03 mW	0.54 mW

Our previous quaternary Vedic multiplier in [17] was designed using many smaller blocks such as current mirrors and current comparators, which constituted the largest part of the multiplier. Hence, only the required blocks were active for any multiplication operation. This had resulted in very low power consumption but at the cost of speed of operation. Since each block must wait for the preceding blocks for its operation, the delay of the multiplier was greater. Besides, a large number of HCCM mirrors had further increased the delay, since such a mirror exhibits greater transition time for the output current in response to the change in input [21]. For the proposed multiplier in this paper, larger functional blocks have been used. Moreover, the number of HCCM mirrors has been greatly reduced. As a result, the worst-case delay has been reduced to 68.12% of that in [17]. The difference between the best-case and worst-case delays is reduced, which indicates that the range of readings of the delay is narrow. In addition, the average of best-case and worst-case delays is approximately equal to the average of all readings of delay, which clearly indicates that the values of delay are centered about the average value. However, power consumption of the proposed multiplier has been increased compared to that in [17].

CONCLUSIONS, DISCUSSION, AND FUTURE SCOPE

The proposed multiplier is found to be area efficient compared to other multipliers in the literature. Transient delay of the proposed multiplier is also improved compared to our previous multiplier in [17]. Supply voltage and unit current step are also very small. Though power consumption is greater, it is still precisely predictable and has values within a confined range. The multiplier proposed in this paper has a regular and hierarchical structure. It works on simple quaternary arithmetic that can be easily understood from the simple truth table provided in this paper. The proposed multiplier does not use any complex mathematical expression like most of the MVL circuits in the literature. It makes the implementation of MVL as simple as binary logic. Moreover, proposed approach allows designing $2^n \times 2^n$ MVL multipliers, where

$n=0,1,2,3$, and so on, making MVL circuits more generalized like binary circuits. The proposed work has achieved a transistor level design of a Vedic multiplier; whereas, most of the Vedic multipliers in the available literature are designed using hardware description languages.

Current-mode MVL results in smaller circuitry and effective utilization of interconnections, whereas, Vedic mathematics provides regular and hierarchical multiplier architecture. The circuit blocks designed in the current-mode MVL, when implemented with Vedic architecture, yield a better multiplier design. Along with the multiplication methods, Vedic mathematics has much more sophisticated mathematical techniques.

Although the proposed multiplier gives a different approach to MVL circuits with some better results, the speed-power performance of the multiplier needs to be improved. This can be done by using high-speed circuit blocks with reduced switching activities and multi-digit adders that are more efficient. In addition, the proposed approach can be used to design a decimal multiplier. This will be the focus of our future work.

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